

Phase 2 SQL Queries

Team: Kislay Jha, Vedansh Shah

1. Easy (E4 - Highly Shared but Poorly Liked)

SQL

```
SELECT post_id, platform, likes, shares, comments
FROM posts
WHERE shares > 1500 AND likes < 500;
```

2. Medium (M2 - Do High Follower Users Get More Engagement?)

SQL

```
WITH UserEngagement AS (
    SELECT
        u.user_id,
        CASE WHEN u.follower_count >= 25000 THEN 'High follower (>= 25k)'
    ELSE 'Low follower (< 25k)' END AS follower_group,
        p.total_engagement
    FROM users u
    JOIN posts p ON u.user_id = p.user_id
)
SELECT
    follower_group,
    COUNT(user_id) AS total_posts,
    ROUND(AVG(total_engagement), 2) AS avg_engagement_per_post
FROM UserEngagement
GROUP BY follower_group;
```

3. Hard (H6 - Find the Most Suspicious High Impact Users)

SQL

```
WITH OverallAvg AS (
    SELECT AVG(total_engagement) AS global_avg FROM posts
),
UserStats AS (
    SELECT
        u.user_id,
        u.location,
        u.follower_count,
        COUNT(p.post_id) AS num_posts,
        SUM(p.total_engagement) AS total_engagement,
        AVG(p.total_engagement) AS avg_engagement,
        SUM(CASE WHEN p.shares > p.likes THEN 1 ELSE 0 END) AS
suspicious_posts_count
    FROM users u
    JOIN posts p ON u.user_id = p.user_id
    GROUP BY u.user_id, u.location, u.follower_count
)
SELECT
    us.user_id,
    us.location,
    us.follower_count,
    us.num_posts,
    ROUND(us.avg_engagement, 2) AS avg_engagement,
    us.total_engagement
FROM UserStats us
```

```
CROSS JOIN OverallAvg oa
WHERE us.follower_count < 10000
      AND us.avg_engagement > oa.global_avg
      AND us.suspicious_posts_count > 0
ORDER BY us.total_engagement DESC;
```